
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

Anonymous authors
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ABSTRACT

Multimodal Large Language Models (MLLMs) incur prohibitive computational overhead on long videos because of the massive number of visual tokens they must process. To improve efficiency, existing methods reduce redundancy by pruning or merging tokens based on importance or similarity. Yet these approaches share a decisive flaw: they ignore what actually drives a video’s meaning, namely its changes and turning points, and they never jointly model spatio-temporal relationships. We resolve this with a clear principle: similarity identifies redundancy, while difference captures key events. Building on this principle, we introduce ST-SimDiff, a training-free framework that operationalizes both sides of this dichotomy. We first construct a spatio-temporal graph over the visual tokens to model their complex associations in a single unified structure. We then apply a parallel dual-selection strategy: 1) similarity-based selection uses community detection to retain representative tokens and aggressively compress static information; 2) temporal difference-based selection pinpoints content-changing points and preserves the tokens that capture key dynamic shifts. Together, these two paths preserve both static and dynamic content with a minimal token budget. Extensive experiments demonstrate that our method decisively outperforms state-of-the-art approaches while sharply reducing computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rise of Large Language Models (LLMs) has propelled Large Vision-Language Models (LVLMs), which now demonstrate remarkable capabilities in image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). For video processing, current LVLMs sample a video into a sequence of frames and convert each frame into hundreds or even thousands of visual tokens for the LLM (Bai et al., 2025; Li et al., 2024). Effective as this paradigm is, it carries a fundamental scalability problem: the number of tokens explodes with video duration and resolution, imposing prohibitive computational and storage burdens that severely limit LVLMs in exactly the settings where they matter most, such as long-video analysis and real-time interaction (Fu et al., 2024b).

To address this challenge, a range of methods have been proposed to enhance the efficiency of LVLMs, and they fall into two categories. The first is importance-based pruning. FastV (Chen et al., 2024) showed that the attention distribution across layers is highly redundant when LVLMs process visual information, so visual tokens with low attention scores in the deeper layers can be pruned to cut computation. The second is similarity-based merging/selection, which exploits the strong similarity among visual tokens both between adjacent video frames and within a single frame. FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from adjacent frames, while VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision encoder level.

Despite their success at visual token reduction, existing methods suffer from two fundamental limitations. First, they treat spatial correlations within a frame and temporal correlations at fixed positions across frames in isolation; without joint spatio-temporal modeling, they cannot capture the complex dynamic events that pervade real video. Second, and more critically, they share a common blind spot: by attending only to informational commonalities such as similarity and importance, they ne-

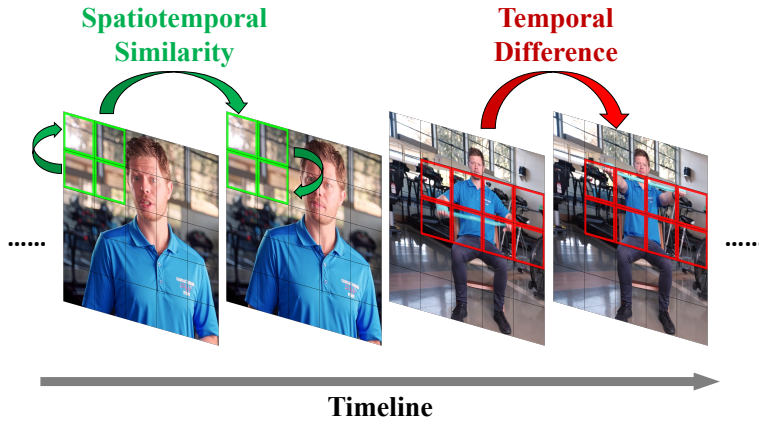


Figure 1: The core motivation of ST-SimDiff. Efficient video understanding demands that two scenarios be handled at once: Spatiotemporal Similarity (left) is exploited to compress redundant information both spatially within a frame and temporally across adjacent frames, while Temporal Difference (right) is detected to capture the key actions and events that define the plot.

glect the changes and differences that actually carry meaning. Because a video’s narrative is driven by turning events, a model that chases only similarity inevitably smooths over sudden actions and misreads the content.

This motivates our central perspective: similarity is for identifying redundancy, while difference is for capturing key moments. Any effective video token compression algorithm must therefore achieve two goals at once: represent the stable content of a video with the fewest possible tokens, and precisely preserve its key changes. We meet both goals with **ST-SimDiff**, a dual token selection framework built on a spatio-temporal graph. The framework has two parts. The first is **similarity-based representative token selection**: we treat all visual tokens of a video as nodes and connect them by feature similarity in the spatio-temporal dimension. A graph community detection algorithm exposes tightly connected token clusters that correspond to stable, persistent visual elements, and from each cluster we keep only a few high-centrality tokens as representatives. The second is **difference-based event token selection**: we focus on the edges along the time axis, and wherever the similarity between corresponding tokens in adjacent frames drops sharply, we flag a turning point and retain those tokens as critical event tokens. We finally merge the two sets and feed them to the LLM. Our contributions can be summarized as follows:

- We are the first to establish that token similarity and token difference deserve equal standing in VLM efficiency research, reframing token compression as the joint problem of compressing redundancy and preserving change.
- We design a spatio-temporal graph that unifies the complex spatio-temporal relationships among video tokens in a single structure, and on top of it we introduce a novel dual token selection strategy that simultaneously screens for both representative and pivotal event tokens.
- Through extensive experiments on video understanding benchmarks, we show that ST-SimDiff sharply reduces the visual token budget while consistently maintaining, and often improving, performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has driven substantial progress in multimodal understanding, giving rise to a wave of prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023).

These LVLMs convert images or video frames into visual tokens via pre-trained visual encoders and feed them to LLMs, with alignment modules such as MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023) bridging the visual encoder and the LLM. This architecture endows LVLMs with exceptional capabilities across multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in video understanding it already handles longer, higher-resolution, and more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This power comes at a steep computational price, most acutely in video understanding. Processing continuous sequences of frames makes the number of visual tokens grow explosively, far beyond that of static images, and directly throttles the training and inference efficiency of LVLMs. Efficiently processing massive video visual tokens and cutting this overhead is therefore a central open problem in current LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Driven by the substantial overhead of video processing, visual token compression has emerged as the key lever for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods approach compression from two angles: token importance and similarity.

Earlier methods focused on importance-based token pruning, identifying and removing visual tokens that contribute little to model performance. They quantify token importance from attention scores or other feature metrics, as in FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Later works (Shang et al., 2024; Ye et al., 2025) additionally exploit token similarity to reduce redundancy more effectively while preserving critical information, recognizing that even important tokens can be highly similar and thus informationally redundant. FrameFusion (Fu et al., 2024b), for instance, combines similarity-based merging with importance-based pruning; VisionZip (Yang et al., 2025) selects dominant tokens and merges contextual information; and VisPruner (Zhang et al., 2025b) leverages visual cues to remove redundant items.

Despite this progress on visual token redundancy, existing work leaves the intricate, global spatio-temporal similarity relationships between tokens largely untouched. In video, visual information is redundant both spatially and temporally, yet similarity-driven methods such as FrameFusion exploit only the temporal similarity between adjacent frames and ignore the richer spatio-temporal structure. This narrow view prevents them from fully exploiting the redundancy inherent in video and sharply limits their performance under extreme token reduction, precisely the regime our work targets.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This step is computationally demanding, because the self-attention mechanism scales quadratically with the token count N , i.e., $O(N^2)$. Our task is to design an efficient token selection function $f(\cdot)$ that maps the full sequence T_{full} and a compression ratio r to a much smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$, maximizing the LVLM’s downstream task performance subject to the constraint that its size $|T_{\text{sub}}|$ is approximately $r \cdot N$. The core challenge, then, is to build a selection function $f(\cdot)$ that retains the most critical information within the subset T_{sub} — and it is exactly this challenge that ST-SimDiff is designed to solve.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff comprises two components that operate in parallel on a spatio-temporal graph built from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The SRTS module identifies and condenses the video’s stable, redundant content. Applying community detection to the graph groups highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and from each cluster we keep a few highly central tokens to form the representative set T_{rep} . The DETS module captures the video’s crucial dynamic shifts. It analyzes the temporal edges

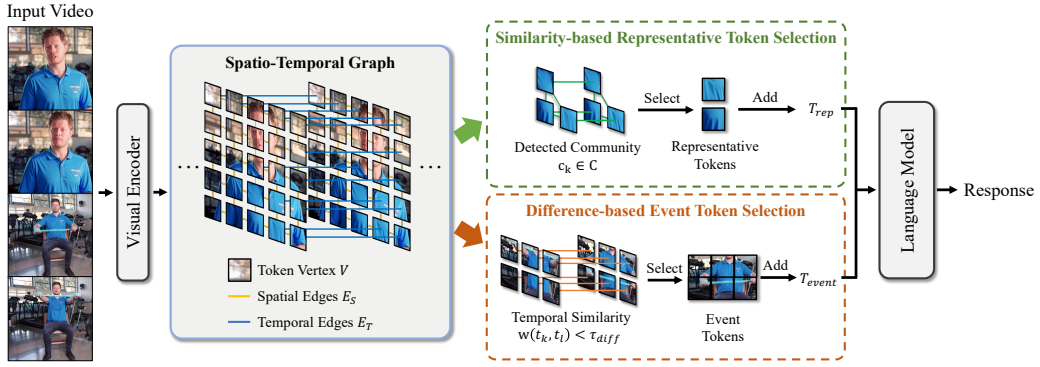


Figure 2: Overview of the ST-SimDiff framework, built from three tightly integrated parts: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

of the graph and pinpoints moments where the similarity between corresponding tokens in adjacent frames drops sharply. These high-difference points mark turning points, and the tokens framing each transition are preserved as critical event tokens, forming the set T_{trans} . Combining these two complementary components, ST-SimDiff produces a final token subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ that is at once highly compact and information-rich, retaining stable content while precisely capturing key events.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a specific spatio-temporal location and is associated with a feature vector $\mathbf{x}_k$. We define the location of each token t_k by its frame index $T(t_k)$, its spatial height index $H(t_k)$, and its spatial width index $W(t_k)$.

We model the relationships between these tokens using a spatio-temporal graph $G = (V, E)$, where the vertex set V represents all tokens in T . The edge set E is the union of two distinct subsets, $E = E_S \cup E_T$, which capture spatial and temporal relationships respectively. The spatial edges E_S connects tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connects tokens that are at the same spatial position but in adjacent frames, capturing temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of any edge $w(v_i, v_j) \in E$ is defined by the cosine similarity of the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph structure efficiently encodes both local spatial relationships and temporal continuity, and it is precisely this unified representation that enables our dual-path token selection strategy.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is saturated with spatio-temporal redundancy. In a static scene that persists for several seconds, the background tokens remain highly similar over time; likewise, an object moving across the screen keeps a high degree of semantic correlation across its tokens even as their spatial positions change. Existing methods handle temporal and spatial similarity in isolation and thus miss half of

this structure. Our spatio-temporal graph, by contrast, captures both jointly: semantically related tokens form dense “communities” or “clusters” based on feature similarity, regardless of their specific frame or location. Selecting representatives from each community is therefore a highly efficient way to compress redundant information while preserving core semantics.

To accurately identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only the edges with weights above this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its internal tokens based on their centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its average similarity with all other tokens within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We then set the intra-community retention rate to the globally preset compression ratio r and keep the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores in each community c_k . This uniform rule handles every community gracefully, including single-node communities of unique tokens, where the sole node is always retained. Formally, the set of representative tokens T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the k highest-scoring elements of a given set. In short, this strategy compresses the static and persistent content of a video efficiently by keeping only the most central tokens of each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, difference defines its “events”. A model that attends only to similarity may capture “what is” but is bound to miss “what happened”. A video’s plot is driven by key events, and the essence of every event is change: a new object appears, an action begins, or a scene transitions. In our model, such changes surface as a sharp difference between the features of temporally adjacent tokens. Capturing these “turning points” of significant difference is therefore indispensable for understanding a video’s dynamic content and answering “when” and “why” questions correctly.

Difference along the temporal dimension is a reliable signal of a key event: the entry of a new object, a scene change, or the start or end of an action produces drastic changes in the visual features at corresponding positions across adjacent frames. We therefore analyze the temporal edges (E_T) of our spatio-temporal graph directly. For any temporal edge $(v_k, v_l) \in E_T$ connecting two temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). Whenever the difference score of a temporal edge exceeds this threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, we mark the subsequent token t_l as a critical event token and retain it. Formally, the set of event tokens T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This strategy preserves exactly those tokens that signify moments of significant temporal change. By capturing these key transitions, we guarantee that the crucial dynamic aspects of a video’s narrative survive compression rather than being smoothed away.

4.5 OVERALL REDUCTION PROCESS

In our framework, we compute the representative token set T_{rep} and the event token set T_{event} in parallel and take their union $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$ as the initial candidate set. To hit the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$ exactly, we add a final pruning step: whenever $|T_{\text{candidate}}|$ exceeds N_{target} , we drop the $|T_{\text{candidate}}| - N_{\text{target}}$ least important tokens, where importance follows existing work (Chen

et al., 2024) and is read off from the attention scores in the shallow layers of the LLM. This step lets our method meet any computational budget while keeping the most critical information.

In summary, our method unites two complementary strategies: graph-based token selection and attention-based dynamic pruning. The former exploits the intrinsic structure of video content (similarity and difference) to guarantee that core information is retained, while the latter flexibly enforces any given computational budget. This design makes our token compression process both principled and adaptive, striking a robust balance between efficiency and performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff decomposes into three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS). We show that all three run in linear time.

The first stage builds the graph. For each of the N visual tokens, we compute the cosine similarity with its spatially and temporally adjacent neighbors. Since each token has a small, constant number of neighbors, this is a single pass over all tokens with complexity $O(Nd)$, where d is the feature dimension of each token.

The SRTS module then identifies token clusters. As detailed in our implementation (Section 5.2), we employ a connected components algorithm with near-linear time complexity $O(N + |E|)$; because the number of edges $|E|$ is at most $3N$, this reduces to $O(N)$. The subsequent per-community filtering could in principle be costly — a naive all-pairs similarity within each community c_k would cost $O(|C| \cdot |c_k|^2 \cdot d)$ — so we cap the maximum community size and, in the rare case that a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), partition it. This keeps filtering efficient and prevents it from dominating the overall complexity. Mainstream graph clustering algorithms such as the Louvain and Leiden methods offer richer community definitions but at a higher cost approaching $O(N \log N)$; we therefore deliberately adopt the simpler, highly efficient connected components algorithm to minimize processing overhead.

The DETS module makes a single pass over all temporal edges, comparing token similarities against a threshold to flag significant changes, again at linear complexity $O(Nd)$.

Taken together, these linear-time operations give ST-SimDiff an overall complexity of $O(Nd)$ — dramatically cheaper than the quadratic $O(N^2d)$ self-attention of the Large Language Model. Our method thus delivers substantial token reduction at a computational cost that is negligible next to the model’s main inference.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate our method rigorously, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and pit our approach against a comprehensive set of state-of-the-art video token compression techniques. These baselines span the full spectrum of strategies: **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For a fair and complete comparison, we additionally bracket the results with the original uncompressed model (Vanilla) as an upper bound and Random Sampling of an equivalent number of tokens as a lower bound.

Benchmarks To stress-test our method on long-video understanding, we adopt three of the most challenging benchmark datasets available: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a comprehensive benchmark for foundational video understanding, spanning diverse video types and three length categories: Short, Medium, and Long. LongVideoBench probes the ability to retrieve and reason about details in long videos through a “referring reasoning” task, with videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset over long-term, first-person videos that directly tests a model’s grasp of character intentions and causal chains of actions.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video understanding benchmarks on LLaVA-Video-7B for 64 input frames (%). The best performance in each column is emphasized in **bold**, and our method attains it on nearly all of them.

5.2 IMPLEMENTATION DETAILS

We run on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04, and conduct all inference evaluation with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video we follow its original setting of 64 input frames, and for a fair comparison of computational efficiency and resource usage we hold NVILA to the same 64-frame input. We fix the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$, and report results at token compression rates of $r = 30\%$ and $r = 50\%$. In practice, balancing time consumption against performance, we use connected component finding as our community detection algorithm.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We evaluate ST-SimDiff against a broad set of mainstream efficient video language model (VLM) compression methods on three widely-used long-form video understanding benchmarks: VideoMME, LongVideoBench, and EgoSchema. To demonstrate that its gains generalize and are robust, we report detailed results on two distinct base models, LLaVA-Video-7B and NVILA-8B, under token retain ratios of 30% and 50%. The results appear in Table 1 and Table 2.

(1) ST-SimDiff consistently delivers the best overall results across every test configuration. Most strikingly, at a 50% token retain ratio the overall performance of our method on both base models not only surpasses all other compression algorithms but, on several benchmarks, matches or exceeds the original model that uses 100% of the tokens — that is, we compress the token budget by half at no cost to accuracy. This outcome confirms both the effectiveness and the novelty of ST-SimDiff and motivates a closer look at the strategies and limitations of existing baselines.

(2) Among importance-based baselines (e.g., FastV, FasterVLM), FasterVLM is the stronger performer, yet all such methods share a decisive weakness: they handle the temporal redundancy pervasive in video poorly, retaining important but repetitive tokens and thereby wasting compression budget. Hybrid baselines (e.g., PruMerge, VisionZip, FrameFusion) are more competitive, with

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video understanding benchmarks on NVILA-Video-8B for 64 input frames (%). The best performance in each column is emphasized in **bold**, and our method claims it on nearly all of them.

FrameFusion the strongest of them overall. Even so, they all optimize for the same objective — preserving representative “commonality” information — and therefore inherit a structural blind spot: they overlook the key narrative information carried by “turning points” and “abrupt events”. This is precisely the gap ST-SimDiff is built to close.

(3) In short, existing baselines embody a steadily deepening understanding of “redundancy” in video, but they remain constrained by two core limitations: they lack any synergistic analysis of spatio-temporal joint similarity between visual tokens, so they cannot capture complex redundancies; and they neglect “difference” detection, so they fail to preserve the key changes that define plot development. ST-SimDiff resolves both at once — its spatio-temporal graph uniformly models the complex spatio-temporal relationships that baselines miss, and its difference detection recovers exactly the narrative information they discard — and this is what drives its consistent performance gains.

5.4 ABLATION STUDY

To isolate the contribution of each core component of ST-SimDiff, we ran a series of detailed ablation studies. Starting from a Baseline that uses only fundamental importance-based pruning, we progressively added our similarity selection module (+Sim) and difference selection module (++Diff) to quantify exactly what each contributes. All experiments use token retain ratios of 30% and 50%, with results in Table 5.

(1) We first evaluate the +Sim module. Adding similarity modeling on top of the baseline improves performance consistently across every benchmark and ratio. Breaking the similarity strategy into spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) variants makes the reason plain: jointly modeling the spatio-temporal associations between visual tokens is decisively the most effective way to compress redundancy, outperforming either axis alone.

(2) On top of the optimal spatio-temporal joint similarity module, we add the difference selection module (++Diff) to complete the framework. This module produces a clear performance leap, most

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Ablation results of different components (%). Each module adds a clear, consistent gain, and the full framework is best throughout.

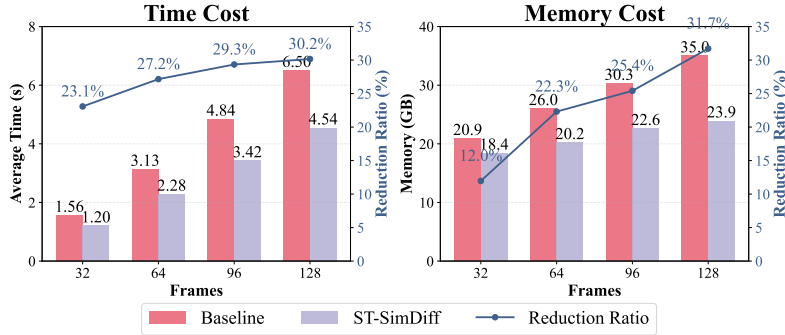


Figure 3: Computational cost comparison between our method and the baseline LLaVA-Video model in average inference time and peak GPU memory usage under a 30% token budget. ST-SimDiff’s advantage widens as videos get longer.

pronounced under the aggressive 30% compression ratio, and demonstrates its indispensable complementary role: where the +Sim module compresses redundancy, the ++Diff module reliably identifies and preserves the unique event tokens that carry the video’s narrative. Its marginal gain naturally shrinks at 50% retention, simply because the more lenient +Sim module has already captured many of these event tokens and performance is nearing its upper bound. This analysis directly confirms our core motivation: the framework’s strength is its ability to balance similarity-based redundancy compression against the essential preservation of difference-based events, and that balance pays off most under demanding high-compression conditions.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify ST-SimDiff’s practical efficiency, we measured its inference time and peak GPU memory usage under a 30% token budget and compared it against the uncompressed baseline (Baseline LLaVA-Video) across video inputs from 32 to 128 frames. The results in Figure 3 make the efficiency advantage unmistakable.

Inference Time ST-SimDiff delivers a substantial and growing speedup. As the number of input frames rises, the baseline’s average inference time climbs sharply from 1.56 seconds (32 frames) to 6.50 seconds (128 frames), whereas our method holds this to just 4.54 seconds at 128 frames.

Crucially, the time saved scales with video length, rising from 23.0% at 32 frames to 30.2% at 128 frames — the longer the video, the more ST-SimDiff wins.

Memory Cost ST-SimDiff is equally strong on memory. The baseline’s peak GPU memory grows linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames). By sharply cutting the number of tokens fed to the large language model, our method relieves this pressure decisively: on 128-frame videos its peak memory is only 23.9 GB, a 31.7% reduction over the baseline. This directly enables longer videos to be processed on memory-constrained hardware, substantially broadening where our model can be deployed.

In summary, ST-SimDiff achieves leading task performance and, at the same time, decisive advantages in both inference time and memory — a combination that establishes it as a genuinely efficient solution for video understanding.

6 CONCLUSION

In this paper, we tackle the computational inefficiency of Multimodal Large Language Models on long videos — a problem rooted in the excessive number of visual tokens and, just as importantly, the neglect of critical content changes. We introduce ST-SimDiff, a training-free framework that builds a spatiotemporal graph to model the complex associations between tokens and applies a dual selection strategy: similarity-based selection identifies redundancy via community detection, while temporal difference-based selection preserves tokens at key transition points. Extensive experiments on VideoMME and LongVideoBench show that ST-SimDiff sharply reduces processing time and memory usage while consistently surpassing state-of-the-art performance. More broadly, by giving equal weight to content representativeness and transitional events, this work opens a new research direction toward more efficient and more faithful representations of video.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

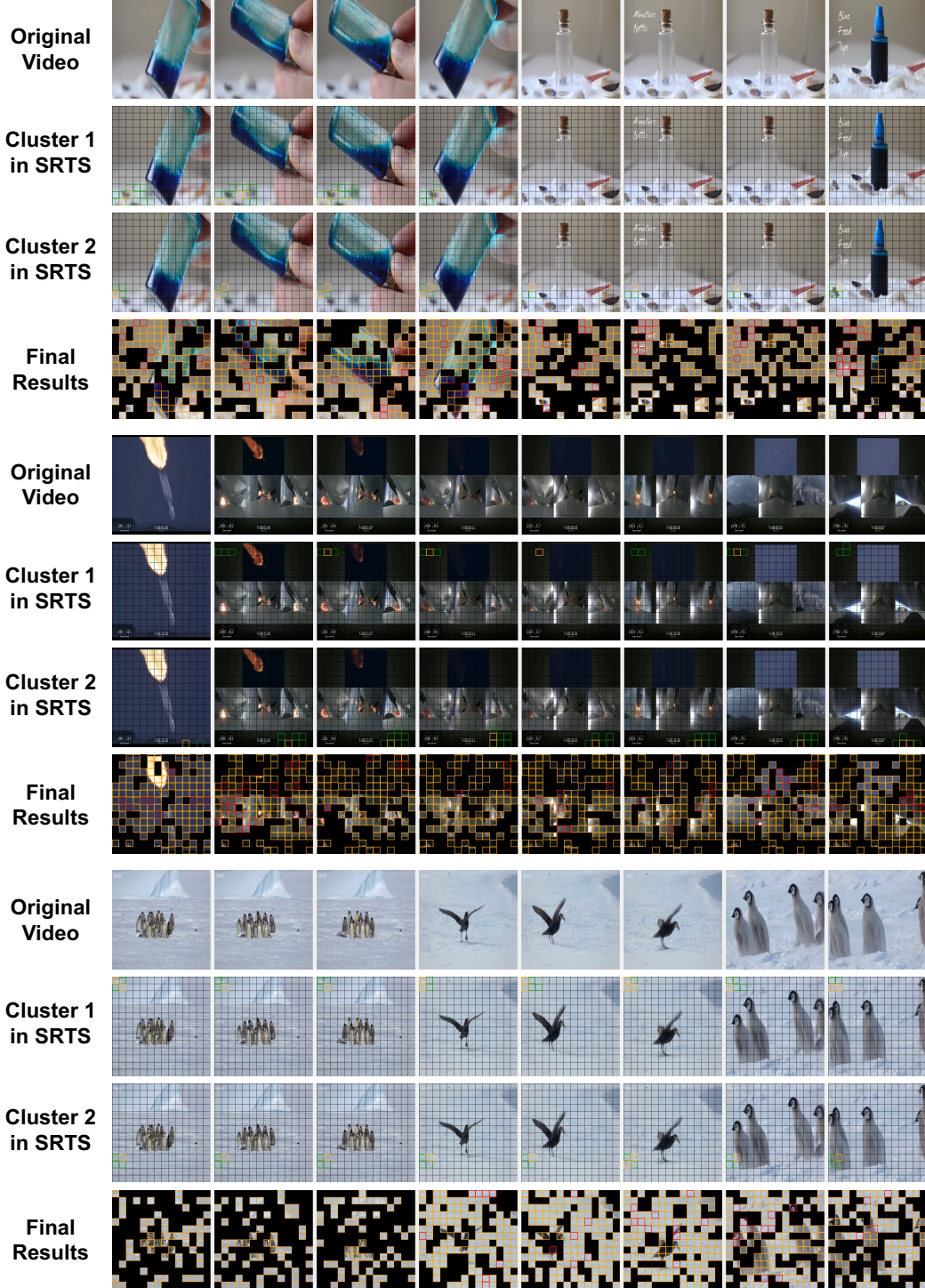


Figure 4: Visualization of the ST-SimDiff process. The token selection is decomposed into the communities detected by SRTS (shown as grids in Cluster 1 and 2) and the event-driven tokens detected by DETS. The final result lays bare the synergy at the heart of our design: sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

To make the ST-SimDiff framework concrete, Figure 4 presents visualization samples of the intermediate and final token selection results on a sample video sequence featuring a dynamic object manipulation task. For the Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show our graph community detection algorithm at work: it cleanly groups spatially and temporally redundant regions, such as the static background and stationary objects, into cohesive communities. This confirms that SRTS reliably identifies stable visual elements and compresses them down to a few central “representative tokens,” shown as sparse yellow boxes in the final results — exactly the mechanism that lets the model absorb massive background redundancy at almost no cost.

Complementing this, the Difference-based Event Token Selection (DETS) takes over during dynamic moments. In the “Final Results” row, tokens triggered by high temporal difference are marked with dense red bounding boxes that align precisely with the moving hand and the shifting blue liquid, verifying that our method pinpoints the “turning points” where the temporal difference exceeds the threshold (τ_{diff}). This is what preserves fine-grained action details and rapid visual changes, letting the model capture the essence of the event instead of drowning in static data.

Finally, the synergy of the two components is unmistakable in the overlay of the final results, which exhibits a powerful balance: sparse yellow tokens carry the stable background, while dense red tokens capture the rapid motion. This contrast is direct visual evidence for our core motivation. ST-SimDiff sidesteps the blind spots of similarity-only approaches by sampling densely during rapid changes, and it simultaneously prunes the spatiotemporal redundancies that importance-based methods leave behind. The result is an optimal trade-off between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video understanding benchmarks on Qwen2.5-VL-7B for 64 and 128 input frames (%). At a 50% budget, ST-SimDiff essentially recovers the full-token upper bound.

To further establish that our framework generalizes beyond a single architecture, we conducted additional experiments on the recently released Qwen2.5-VL model, evaluating on the VideoMME and LongVideoBench benchmarks.

The results in Table 4 confirm that ST-SimDiff transfers cleanly to Qwen2.5-VL. At a 30% token retention ratio, our method reaches 62.4 on VideoMME and 56.8 on LongVideoBench; raising the budget to 50% lifts these to 62.9 and 58.4, respectively. Most tellingly, at the 50% ratio the VideoMME score matches the upper bound obtained with all tokens, and the LongVideoBench score sits right alongside its full-performance counterpart. Together these findings make the generalization ability of our framework unmistakable.

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Ablation results of different components on the NVILA-Video-8B model (%). The gains mirror those on LLaVA-Video, with the full framework best at both ratios.

C ABLATION STUDY RESULTS ON NVILA

To confirm that each component generalizes across base models, we repeated our detailed ablation on the NVILA-Video-8B model (Liu et al., 2024c). The setup matches the ablation in the main paper: starting from a Baseline that uses only fundamental importance-based pruning, we progressively introduce the similarity selection module (+ Sim) and the difference selection module (++) Diff).

As Table 5 shows, the results reaffirm our design and track the LLaVA-Video-7B trends closely. Adding the similarity module (+ Sim) on top of the baseline yields a steady improvement at both 30% and 50% token retain ratios. Within it, we compared spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) strategies, and the joint strategy is clearly the most effective: at $r = 50\%$, for instance, it reaches 61.4% on VideoMME, beating both single-axis variants. This once again establishes that jointly modeling spatio-temporal associations is essential for identifying redundant information. Adding the difference selection module (++) Diff) on top of the optimal joint similarity module completes our ST-SimDiff framework, and the table shows it delivers the largest performance gains of all. This is compelling evidence for the effectiveness and generalization of our two core modules — similarity and difference selection — and directly validates our central idea of handling redundancy and change together.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

The similarity threshold τ_{sim} and the difference threshold τ_{diff} are the two key hyperparameters of our framework, governing the selection of representative and transitional tokens respectively. Figure 6 reports their ablation. Both parameters follow the same rise-then-fall trend and, importantly, remain robust across a wide range. Setting τ_{sim} too low blurs community detection, while setting it too high fragments semantic clusters; setting τ_{diff} too low admits noise through over-sensitivity, while setting it too high causes key events to be missed. The model is optimal at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, and we adopt this configuration throughout our other experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

To confirm that our key hyperparameters are equally robust on a second architecture, we repeated the ablation on the NVILA-Video-8B model. As shown in Fig. 6, we evaluated the impact of the

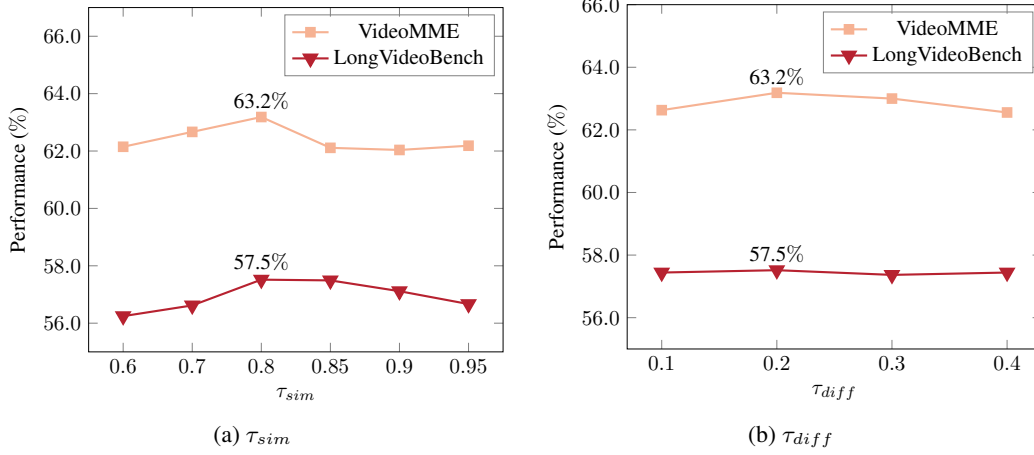


Figure 5: Ablation study results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench. Performance stays stable over a wide range, peaking at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$.

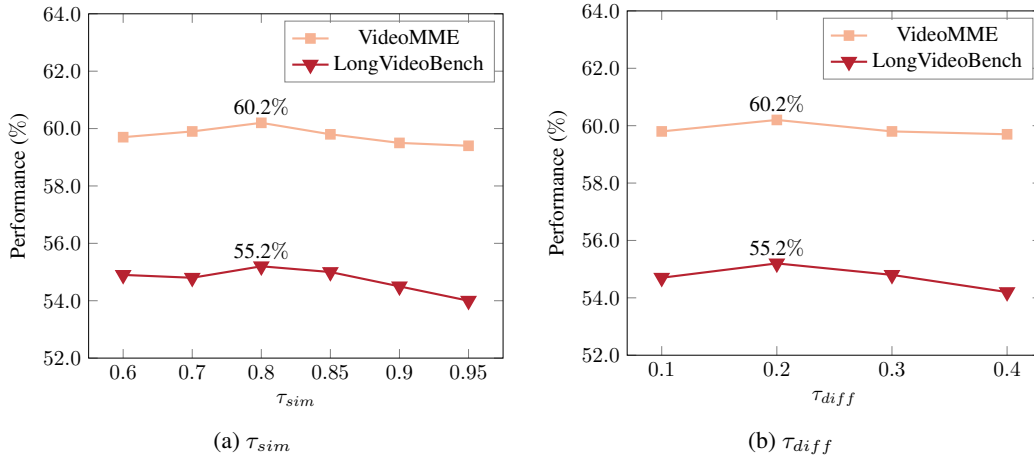


Figure 6: Ablation study results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B. The same stable, single-peak behavior holds, again favoring $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$.

similarity threshold τ_{sim} and the difference threshold τ_{diff} on performance across VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We swept τ_{sim} over $[0.6, 0.95]$. Performance is remarkably insensitive to this threshold: VideoMME peaks at 60.2% and LongVideoBench at 55.2%, both at $\tau_{\text{sim}} = 0.8$. Values that are too high or too low cost only a small amount, so the overall fluctuation is minimal — clear evidence that our method is stable across a wide range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . We swept τ_{diff} over $[0.1, 0.4]$. As with τ_{sim} , performance is strongly robust to this threshold, peaking around $\tau_{\text{diff}} = 0.2$ on both benchmarks — 60.2% on VideoMME and 55.2% on LongVideoBench. Setting the difference threshold near 0.2 is therefore the most effective choice for capturing key events, maximizing performance while compressing tokens.

In summary, these studies establish that our method is robust to its key hyperparameters and firmly justify the choice of $\tau_{\text{sim}} = 0.8$ and $\tau_{\text{diff}} = 0.2$ in the main experiments, since these values consistently yield the best performance across benchmarks and base models alike.